DO NOT WRITE IN THIS AREA

21 (a) Use your calculator to work out $\dfrac{\sqrt{8.7^2 + 12.4}}{9.5 - 1.2^3}$

Write down all the figures on your calculator display.

$8.7^2 + 12.4 = 88.09$; $\sqrt{88.09} = 9.385627311...$

$9.5 - 1.2^3 = 9.5 - 1.728 = 7.772$

$9.385627311... \div 7.772 = 1.207620601$

2 marks total

Evaluate numerator and denominator; keep full precision.

1.207620601

(2)

(b) Write your answer to part (a) correct to 3 significant figures.

$1.207620601 \rightarrow 1.21$ (3 s.f.)

1.21

(1)

1 mark

The fourth significant digit is 7, so round up.

(Total for Question 21 is 3 marks)